



Original research article

The spherical linear canonical transform: Definition and properties

Hui Zhao, Bing-Zhao Li^{*}

School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China
 Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China

ARTICLE INFO

Keywords:

Spherical Fourier transform
 Spherical linear canonical transform
 Spherical linear canonical Hankel transform
 Convolution theorems
 Spherical polar coordinates

ABSTRACT

The spherical Fourier transform has attracted considerable attention in the fields of acoustics, optics, and heat because of its superiority in solving practical problems- within the confines of spherical symmetry. A spherical linear canonical transform in spherical polar coordinates is investigated in this study. First, definitions of the spherical linear canonical transform and spherical linear canonical Hankel transform are proposed. Second, the relationship between the spherical linear canonical transform and spherical linear canonical Hankel transform is derived based on the orthogonality of the spherical harmonics. Finally, several essential properties of the proposed spherical linear canonical transform were obtained based on this relationship, including linearity, inversion formulas, shifts, and convolution theorems. Finally, potential applications of the spherical linear canonical transform are discussed.

1. Introduction

In the field of modern signal processing, the search for novel signal processing theories and methods continues to attract considerable research attention. With the increasing demand for signal characteristics in modern information and communication fields, a series of signal processing transformation methods have been proposed based on classical Fourier transform (FT), such as fractional Fourier transform (FRFT), short-time FRFT, wavelet transform, fractional Stockwell transform, and linear canonical Stockwell transform [1–5]. These transformation methods were studied using the traditional Cartesian coordinate system. Although they have their advantages, they cannot achieve optimal results when dealing with complex problems with spherical symmetry properties such as the three-dimensional location of a sound source, thermal imaging, and paraxial optical systems [6–9]. To solve these problems, researchers have proposed a new signal transformation method called the spherical Fourier transform (SFT) [10,11].

The SFT is a new form of transformation for the three-dimensional FT in spherical polar coordinates, which is most useful when the function being transformed has some underlying spherical symmetry. It is widely used in the fields of acoustics, optics, and astronomy. Tang et al. proposed a method for spatial localization of a sound source, where SFT was employed to transform the sound field into the wave vector-frequency domain to obtain the azimuthal information of the sound sources [12]. Using the SFT theory, a non-conformal sound field transformation of the transfer matrix based on spherical harmonic decomposition was proposed [13], and the theoretical expression of the method was deduced. Similarly, SFT has important applications in thermal imaging [9,14].

The objects in the aforementioned problems are stationary signals. SFT reflects the overall characteristics of the signal and cannot be described as a local time change characteristic of non-stationary signals. In practical applications, the signals can be complex, with time-varying and non-stationary characteristics. SFT is not efficient when processing non-stationary signals, and it cannot describe the signal frequency change over time, which has certain limitations. Further SFT is not optimal for non-band-limited signal processing on the sphere. Therefore, to overcome these shortcomings of SFT, it is necessary to find a new transformation

^{*} Corresponding author at: Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China
 E-mail address: li_bingzhao@bit.edu.cn (B.-Z. Li).

method for application in different fields. As a generalized form of linear integral transformation, the linear canonical transform (LCT) has a unique advantage in processing non-stationary signals.

LCT belongs to the class of generalized linear integral transforms with three parameters [15,16]. When special parameters are selected, they can be reduced to a series of classical time–frequency transformations. LCT has proven to be a powerful tool for analyzing non-stationary signals. The non-band-limited signals in both traditional FT domains may be band-limited in the LCT domain. Therefore, the LCT-based signal processing tool can compensate for the shortcomings of SFT.

This study proposes a new transformation method, namely the spherical linear canonical transform (SLCT), which is a generalization of a class of multi-parameter linear integral transformations in spherical polar coordinates. The embedding of free parameters facilitates more flexibility in the analysis and processing of non-stationary signals, including representation, convolution theorem, and sampling, compared to that with SFT. The main contributions of this study are summarized as follows.

- Proposing definitions of SLCT and spherical linear canonical Hankel transform (SLCHT) in spherical polar coordinates.
- The relationship between the SLCT and SLCHT is derived based on the orthogonality of the spherical harmonics.
- Some basic properties and convolution theorems of the SLCT in spherical polar coordinates are investigated based on this relationship.

The remainder of the paper is structured as follows: Section 2 provides some basic preliminaries. In Section 3, the definitions of the SLCT and SLCHT in spherical polar coordinates are provided, and the relationship between the SLCT and SLCHT is derived. Based on this relationship, some basic properties are investigated. In Section 4, we present two types of convolution theorems for SLCT. In Section 5, potential applications are presented to demonstrate the importance of the SLCT. Finally, Section 6 concludes the paper.

2. Preliminaries

2.1. Spherical harmonics

The spherical harmonic function is a characteristic function of the Laplace operator on a sphere in the following form [17,18]:

$$Y_{l,m}(\psi, \theta) = \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} P_{l,m}(\cos \psi) e^{im\theta}, \quad \text{for } l \geq 0, -l \leq m \leq l \quad (1)$$

These are spherical harmonics of degree l and order m , where l and m are integers. $P_{l,m}$ is an associated Legendre function, $0 \leq \psi \leq \pi$ represents the colatitude, and $0 \leq \theta \leq 2\pi$ represents the longitude. The spherical harmonic function is the angular part of the spherical coordinate system of the formal solution of the Laplace equation. This is a well-known function in modern mathematics. They are widely used in quantum mechanics, computer graphics, light processing, and spherical mapping.

Spherical harmonics have positive orthogonal properties; that is,

$$\int_0^{2\pi} \int_0^\pi Y_{l,m} \overline{Y_{l',m'}} \sin \psi d\psi d\theta = \delta_{l,l'} \delta_{m,m'}, \quad (2)$$

where $\delta_{i,j}$ is the Kronecker delta and $\overline{Y_{l',m'}}$ stands for the complex conjugate. The normalization of the spherical harmonics may differ. The spherical harmonics form a complete set of orthonormal functions that can form a vector space.

2.2. Spherical fourier transform and spherical Hankel transform

The SFT of any function $f(r, \varphi, \theta)$ is defined as [18]:

$$F(\rho, \alpha, \beta) = \mathcal{F}[f](\rho, \alpha, \beta) = \frac{1}{(2\pi)^3} \int_0^\infty \int_0^{2\pi} \int_0^\pi f(r, \varphi, \theta) e^{-ir\rho \cos \Theta} r^2 \sin \varphi dr d\varphi d\theta, \quad (3)$$

where $\cos \Theta = \sin \varphi \sin \alpha \cos(\theta - \beta) + \cos \varphi \cos \alpha$. It must be emphasized that $\cos \Theta$ mainly refers to the writing [19,20], and is only a defining symbol for the convenience of the following proof. The inverse SFT operator is denoted by $\mathcal{F}^{-1}$.

The spherical Hankel transform (SHT) can be defined as [18,21,22]:

$$\hat{F}_n(\rho) = \mathbb{S}_n\{f(r)\} = \int_0^\infty f(r) j_n(\rho r) r^2 dr, \quad (4)$$

where $\mathbb{S}_n$ is used to specifically denotes the SHT of order n , and the spherical Bessel function $j_n(z)$ is generally defined by the spherical Bessel equation [18,22]:

$$j_n(z) = \sqrt{\frac{\pi}{2z}} J_{n+1/2}(z), \quad (5)$$

where $J_{n+1/2}(z)$ is a Bessel function of half-integer order.

The inverse transform is given by [18]:

$$f(r) = \frac{2}{\pi} \int_0^\infty \hat{F}_n(\rho) j_n(\rho r) \rho^2 d\rho. \quad (6)$$

2.3. Linear canonical transform

The three-dimensional linear canonical transform of a signal $f(t) = f(x, y, z)$ with real parameter $A = (a, b \neq 0; c, d)$ is defined as [15,23]

$$F^A(\mathbf{u}) = F^A(u_x, u_y, u_z) = \frac{1}{(\sqrt{2\pi ib})^3} \int_{\mathbb{R}^3} f(\mathbf{t}) e^{i\left(\frac{a}{2b}|\mathbf{t}|^2 + \frac{d}{2b}|\mathbf{u}|^2 - \frac{1}{b}\mathbf{t} \cdot \mathbf{u}\right)} d\mathbf{t} \quad (7)$$

where $\mathbf{t} = (x, y, z)$, $\mathbf{u} = (u_x, u_y, u_z)$, $|\mathbf{t}|^2 = x^2 + y^2 + z^2$, $\mathbf{t} \cdot \mathbf{u} = xu_x + yu_y + zu_z$, and $|\mathbf{u}|^2 = u_x^2 + u_y^2 + u_z^2$.

The corresponding inverse formula is given by:

$$f(\mathbf{t}) = \frac{1}{(\sqrt{-2\pi ib})^3} \int_{\mathbb{R}^3} F^A(\mathbf{u}) e^{-i\left(\frac{a}{2b}|\mathbf{t}|^2 + \frac{d}{2b}|\mathbf{u}|^2 - \frac{1}{b}\mathbf{t} \cdot \mathbf{u}\right)} d\mathbf{u}. \quad (8)$$

3. SLCT and SLCHT in spherical polar coordinates

3.1. Definitions

By the usual coordinate transformation from Cartesian coordinates to spherical polar coordinates, the spatial position $\mathbf{t} = (x, y, z) = (r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi)$, where $0 \leq \varphi \leq \pi$, $0 \leq \theta \leq 2\pi$, and r is the radial distance from the origin. Similarly, the frequency space variable is transformed into a spherical polar coordinate system $\mathbf{u} = (u_x, u_y, u_z) = (\rho \sin \alpha \cos \beta, \rho \sin \alpha \sin \beta, \rho \cos \alpha)$. Hence, we obtained the expression of SLCT in spherical polar coordinates.

Definition 1 (SLCT). The spherical linear canonical transform (SLCT) of $f(r, \varphi, \theta)$ with real parameter $A = (a, b \neq 0; c, d)$ in spherical polar coordinates is defined as:

$$F^A(\rho, \alpha, \beta) = \mathcal{L}_A[f](\rho, \alpha, \beta) = \int_0^\infty \int_0^{2\pi} \int_0^\pi f(r, \varphi, \theta) K_A(\rho, \alpha, \beta; r, \varphi, \theta) r^2 \sin \varphi dr d\varphi d\theta, \quad (9)$$

where K_A denotes the SLCT kernel,

$$K_A(\rho, \alpha, \beta; r, \varphi, \theta) = \frac{1}{(\sqrt{2\pi ib})^3} e^{i\left(\frac{d}{2b}\rho^2 - \frac{1}{b}r\rho \cos \Theta + \frac{a}{2b}r^2\right)}, \quad (10)$$

where a, b, c, d are real numbers satisfying $ad - bc = 1$, r is the radial distance from the origin, $0 \leq \varphi, \alpha \leq \pi$, $0 \leq \theta, \beta \leq 2\pi$, $\cos \Theta = \sin \varphi \sin \alpha \cos(\theta - \beta) + \cos \varphi \cos \alpha$. Similarly, $\cos \Theta$ mainly refers to the writing [19,20], and is only a defining symbol for convenience of the following proof.

Remark 1. When $A = (0, 1; -1, 0)$, SLCT can be reduced to SFT. It indicates a relationship between SLCT and SFT

$$F^A(\rho, \alpha, \beta) = \frac{4\pi^2}{\sqrt{-2\pi ib^3}} e^{i\frac{d}{2b}\rho^2} F[\tilde{f}]\left(\frac{\rho}{b}, \alpha, \beta\right), \quad (11)$$

where $\tilde{f}(r, \varphi, \theta) = e^{i\frac{a}{2b}r^2} f(r, \varphi, \theta)$.

Next, we define the spherical linear canonical Hankel transform (SLCHT) for spherical coordinates.

Definition 2 (SLCHT). The spherical linear canonical Hankel transform (SLCHT) with real parameter $A = (a, b \neq 0; c, d)$ is defined as:

$$\hat{F}_n^A(\rho) = \mathbb{S}_n^A\{f(r)\} = \frac{e^{i\frac{d}{2b}\rho^2}}{b} \int_0^\infty e^{i\frac{a}{2b}r^2} f(r) j_n\left(\frac{\rho r}{b}\right) r^2 dr, \quad (12)$$

where $\mathbb{S}_n$ denotes the SLCHT of order n , and the spherical Bessel function is given by Eq. (5).

The inversion formula of the SLCHT with $A^{-1} = (d, -b; -c, a)$ takes the form:

$$\begin{aligned} f(r) &= \hat{F}_n^{-A^{-1}}[\hat{F}_n^A[f]](r) \\ &= \frac{e^{-i\frac{a}{2b}r^2}}{b} \int_0^\infty e^{-i\frac{d}{2b}\rho^2} \hat{F}_n^A[f](\rho) j_n\left(\frac{\rho r}{b}\right) \rho^2 d\rho. \end{aligned} \quad (13)$$

Remark 2. When $A = (0, 1; -1, 0)$, SLCHT can be reduced to SHT. This suggests a relationship between SLCHT and SHT defined as:

$$\hat{F}_n^A(\rho) = \frac{e^{i\frac{d}{2b}\rho^2}}{b} \hat{F}_n[\tilde{f}]\left(\frac{\rho}{b}\right), \quad (14)$$

where $\tilde{f}(r) = e^{i\frac{a}{2b}r^2} f(r)$.

3.2. Relationship between the SLCT and the SLCHT

To start from the spherical harmonic series, f and the SLCT kernel function K_A are expanded into the spherical harmonic series. First, any function $f(r, \varphi, \theta)$ can be expanded into a spherical Fourier series as:

$$f(r, \varphi, \theta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l f_{l,m}(r) Y_{l,m}(\varphi, \theta). \quad (15)$$

Then, according to the popular exponential expansion formula [17], we have

$$e^{-\frac{1}{b} i t \cdot u} = e^{-\frac{1}{b} i r \rho \cos \Theta} = 4\pi \sum_{l=0}^{\infty} \sum_{k=-l}^l (-i)^l j_l \left(\frac{\rho r}{b} \right) \overline{Y_{l,k}(\varphi, \theta)} Y_{l,k}(\alpha, \beta). \quad (16)$$

Therefore, the SLCT kernel function K_A is written in the form of spherical harmonics

$$K_A(\rho, \alpha, \beta; r, \varphi, \theta) = \frac{4\pi}{(\sqrt{2\pi i b})^3} \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l e^{j \left(\frac{d}{2b} \rho^2 + \frac{a}{2b} r^2 \right)} j_l \left(\frac{\rho r}{b} \right) \overline{Y_{l,m}(\varphi, \theta)} Y_{l,m}(\alpha, \beta). \quad (17)$$

Eqs. (15) and (17) are substituted into the definition of the SLCT, and by the orthogonality of the spherical harmonic, the relationship between $F^A(u)$ and the l th order SLCHT of $f_{l,m}(r)$ is obtained as:

$$F^A(\rho, \alpha, \beta) = \frac{2}{\sqrt{-2\pi i b}} \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l \hat{F}_{l,m}^A(\rho) Y_{l,m}(\alpha, \beta), \quad (18)$$

where $\hat{F}_{l,m}^A(\rho)$ is the l th order SLCHT of $f_{l,m}(r)$.

The spherical harmonic coefficients of the SLCT of $f(r, \varphi, \theta)$ are expressed as:

$$F_{l,m}^A(\rho) = \frac{2}{\sqrt{-2\pi i b}} (-i)^l \hat{F}_{l,m}^A(\rho) = \frac{2}{\sqrt{-2\pi i b}} (-i)^l \mathbb{S}_l^A \{f_{l,m}(r)\}, \quad (19)$$

that is, $\hat{F}_{l,m}^A(\rho)$ is the l th order SLCHT of $f_{l,m}(r)$; thus, $\hat{F}_{l,m}^A(\rho) = \mathbb{S}_l^A \{f_{l,m}(r) | r \rightarrow \rho\}$.

3.3. Properties

Next, we deduce several basic properties of the SLCT in spherical polar coordinates.

Proposition 1 (Linearity). For any $a_1, a_2 \in \mathbb{R}$, SLCT is a linear operator, namely

$$\mathcal{L}_A [a_1 f + a_2 g] = a_1 \mathcal{L}_A [f] + a_2 \mathcal{L}_A [g]. \quad (20)$$

Proof. This discussion follows directly from the linearity of the integration involved in Definition 1. $\square$

Proposition 2 (Inversion Formula). If the function $f(r, \varphi, \theta)$ has a spherical Fourier expansion given by Eq. (15), then $f(r, \varphi, \theta)$ has the form:

$$f(r, \varphi, \theta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{2(i)^l}{\sqrt{-2\pi i b}} \frac{e^{-i \frac{a}{2b} r^2}}{b} \int_0^{\infty} e^{-i \frac{d}{2b} \rho^2} F_{l,m}^A(\rho) Y_{l,m}(\alpha, \beta) j_l \left(\frac{\rho r}{b} \right) \rho^2 d\rho, \quad (21)$$

where $F_{l,m}^A(\rho)$ denotes the (l, m) th in the expansion of the SLCT of $f(r, \varphi, \theta)$ in spherical polar coordinates.

Proof. SLCT is expanded in spherical harmonics as:

$$F^A(\rho, \alpha, \beta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l F_{l,m}^A(\rho) Y_{l,m}(\alpha, \beta). \quad (22)$$

From Eqs. (15) and (22) and using the orthonormality of spherical harmonics, we obtain

$$f(r, \varphi, \theta) = \frac{2}{\sqrt{-2\pi i b}} \sum_{l=0}^{\infty} \sum_{m=-l}^l (i)^l Y_{l,m}(\alpha, \beta) \left\{ \frac{e^{-i \frac{a}{2b} r^2}}{b} \int_0^{\infty} e^{-i \frac{d}{2b} \rho^2} F_{l,m}^A(\rho) j_l \left(\frac{\rho r}{b} \right) \rho^2 d\rho \right\}. \quad (23)$$

That is

$$f(r, \varphi, \theta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{2}{\sqrt{-j 2\pi b}} (i)^l \left\{ \frac{e^{-i \frac{a}{2b} r^2}}{b} \int_0^{\infty} e^{-i \frac{d}{2b} \rho^2} F_{l,m}^A(\rho) j_l \left(\frac{\rho r}{b} \right) \rho^2 d\rho \right\} Y_{l,m}(\alpha, \beta), \quad (24)$$

which completes the proof. $\square$

Proposition 3 (Spatial Shift Property). Let $\mathbf{t}_0 = (r_0 \sin \varphi_0 \cos \theta_0, r_0 \sin \varphi_0 \sin \theta_0, r_0 \cos \varphi_0)$. Then

$$f(\mathbf{t} - \mathbf{t}_0) = \frac{32\pi}{b^3} \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l Y_{l,m}(\varphi, \theta) e^{-i \frac{a}{2b} (r^2 + r_0^2 - w^2)} \sum_{l'=0}^{\infty} \sum_{m'=-l'}^{l'} \overline{Y_{l',m'}(\varphi_0, \theta_0)} \\ \times \sum_{l''=|l-l'|}^{l+l'} c^{l''}(l, m, l', m') \int_0^{\infty} f_{l'',(m-m')}(w) S_{l''}^{l,l'}(w, r, r_0) \Lambda(r, r_0) w^2 dw, \quad (25)$$

where $\Lambda(r, r_0) = \sum_{l=0}^{\infty} \sum_{m=-l}^l j_l \left(\frac{arr_0}{b} \right) Y_{l,m}(\varphi, \theta) \overline{Y_{l,m}(\varphi_0, \theta_0)}$, $c^{l''}(l, m, l', m')$ is the Slater coefficient [24,25] and $S_{l''}^{l,l'}(w, r, r_0)$ is defined as [26,27]:

$$S_{l''}^{l,l'}(w, r, r_0) = \int_0^{\infty} j_{l''} \left(\frac{\rho w}{b} \right) j_{l'} \left(\frac{\rho r_0}{b} \right) j_l \left(\frac{\rho r}{b} \right) \rho^2 d\rho. \quad (26)$$

Proof. According to the inverse transform formula of the 3D LCT, it follows that:

$$f(\mathbf{t}) = \frac{1}{(\sqrt{-2\pi ib})^3} \int_{\mathbb{R}^3} F^A(\mathbf{u}) e^{-i \left(\frac{a}{2b} |\mathbf{t}|^2 + \frac{d}{2b} |\mathbf{u}|^2 - \frac{1}{b} \mathbf{t} \cdot \mathbf{u} \right)} d\mathbf{u}. \quad (27)$$

Hence, we obtain

$$f(\mathbf{t} - \mathbf{t}_0) = \frac{1}{(\sqrt{-2\pi ib})^3} \int_{\mathbb{R}^3} F^A(\mathbf{u}) e^{-i \left(\frac{a}{2b} |\mathbf{t} - \mathbf{t}_0|^2 + \frac{d}{2b} |\mathbf{u}|^2 \right) + \frac{1}{b} (\mathbf{t} - \mathbf{t}_0) \cdot \mathbf{u}} d\mathbf{u} \\ = \frac{1}{(\sqrt{-2\pi ib})^3} \int_0^{\infty} \int_0^{2\pi} \int_0^{\pi} F^A(\rho, \alpha, \beta) e^{-i \left[\frac{a}{2b} (r^2 - r_0^2) + \frac{d}{2b} \rho^2 \right] + \frac{a}{b} i \mathbf{t} \cdot \mathbf{t}_0 + \frac{1}{b} i (\mathbf{t} - \mathbf{t}_0) \cdot \mathbf{u}} \rho^2 \sin \alpha d\rho d\alpha d\beta. \quad (28)$$

By the popular exponential expansion formula [17,18], we obtain

$$f(\mathbf{t} - \mathbf{t}_0) = \frac{1}{(\sqrt{-2\pi ib})^3} \int_0^{\infty} \int_0^{2\pi} \int_0^{\pi} \frac{2}{\sqrt{-2\pi ib}} \sum_{l''=0}^{\infty} \sum_{m''=-l''}^{l''} (-i)^{l''} \frac{e^{i \frac{d}{2b} \rho^2}}{b} \int_0^{\infty} e^{i \frac{a}{2b} w^2} f_{l'',m''}(w) \\ \times j_{l''} \left(\frac{\rho w}{b} \right) w^2 dw Y_{l'',m''}(\alpha, \beta) e^{-i \left[\frac{a}{2b} (r^2 - r_0^2) + \frac{d}{2b} \rho^2 \right]} 4\pi \sum_{l'''=0}^{\infty} \sum_{m'''=-l'''}^{l'''} (i)^{l'''} j_{l'''} \left(\frac{arr_0}{b} \right) \\ \times Y_{l''',m'''}(\varphi, \theta) \overline{Y_{l''',m'''}(\varphi_0, \theta_0)} 4\pi \sum_{l'=0}^{\infty} \sum_{m'=-l'}^{l'} (-i)^{l'} j_{l'} \left(\frac{\rho r_0}{b} \right) \overline{Y_{l',m'}(\varphi_0, \theta_0)} Y_{l',m'}(\alpha, \beta) \\ \times 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l (i)^l j_l \left(\frac{\rho r}{b} \right) Y_{l,m}(\varphi, \theta) \overline{Y_{l,m}(\alpha, \beta)} \rho^2 \sin \alpha d\rho d\alpha d\beta. \quad (29)$$

We integrate the radial variable to obtain a “shift” operator with a triple Bessel product, defined as [26,27]:

$$S_{l''}^{l,l'}(w, r, r_0) = \int_0^{\infty} j_{l''} \left(\frac{\rho w}{b} \right) j_{l'} \left(\frac{\rho r_0}{b} \right) j_l \left(\frac{\rho r}{b} \right) \rho^2 d\rho, \quad (30)$$

where $S_{l''}^{l,l'}(w, r, r_0) = 0$ unless $|r - r_0| \leq w \leq r + r_0$.

We integrated the angular variable to obtain the Slater coefficient [18,24,25]; thus,

$$f(\mathbf{t} - \mathbf{t}_0) = \frac{32\pi}{b^3} \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l Y_{l,m}(\varphi, \theta) e^{-i \frac{a}{2b} (r^2 + r_0^2 - w^2)} \sum_{l'=0}^{\infty} \sum_{m'=-l'}^{l'} \overline{Y_{l',m'}(\varphi_0, \theta_0)} \\ \times \sum_{l''=|l-l'|}^{l+l'} c^{l''}(l, m, l', m') \int_0^{\infty} f_{l'',(m-m')}(w) S_{l''}^{l,l'}(w, r, r_0) \Lambda(r, r_0) w^2 dw, \quad (31)$$

where $\Lambda(r, r_0) = \sum_{l''=0}^{\infty} \sum_{m''=-l''}^{l''} j_{l''} \left(\frac{arr_0}{b} \right) Y_{l'',m''}(\varphi, \theta) \overline{Y_{l'',m''}(\varphi_0, \theta_0)}$, which completes the proof. $\square$

Proposition 4 (Frequency Shift Property). Let $\mathbf{u}_0 = (\rho_0 \sin \alpha_0 \cos \beta_0, \rho_0 \sin \alpha_0 \sin \beta_0, \rho_0 \cos \alpha_0)$. Then

$$e^{\frac{1}{b} i \mathbf{u}_0 \cdot \mathbf{t}} f(\mathbf{t}) = \frac{8}{b^3} \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l Y_{l,m}(\varphi, \theta) \int_0^{\infty} f_{l,m}(r) S_{l'}^{l,l}(\rho_0, \rho, \rho) \sum_{l'=0}^{\infty} \sum_{m'=-l'}^{l'} Y_{l',m'}(\varphi_0, \theta_0) \overline{Y_{l',m'}(\varphi, \theta)} \rho^2 d\rho, \quad (32)$$

where $S_{l'}^{l,l}(\rho_0, \rho, \rho)$ is given by Eq. (30).

Proof. According to the inverse transform formula of the 3D LCT, it follows that:

$$e^{\frac{1}{b}iu_0t}f(t) = \frac{1}{(\sqrt{-2\pi ib})^3} \int_0^\infty \int_0^{2\pi} \int_0^\pi F^A(\rho, \alpha, \beta) e^{-i\left(\frac{a}{2b}r^2 + \frac{d}{2b}\rho^2\right) + \frac{1}{b}iu_0t + \frac{1}{b}it \cdot u} \rho^2 \sin \alpha d\rho d\alpha d\beta. \quad (33)$$

By the popular exponential expansion formula [17,18] and $F^A(\rho, \alpha, \beta)$ can be expanded in a spherical Fourier series, we obtain

$$\begin{aligned} e^{\frac{1}{b}iu_0t}f(t) &= \frac{1}{(\sqrt{-2\pi ib})^3} \int_0^\infty \int_0^{2\pi} \int_0^\pi \frac{2}{\sqrt{-2\pi ib}} \sum_{l''=0}^\infty \sum_{m''=-l''}^{l''} (-i)^l \frac{e^{i\frac{d}{2b}\rho^2}}{b} \int_0^\infty e^{i\frac{a}{2b}r^2} f_{l'',m''}(r) j_{l''}\left(\frac{\rho r}{b}\right) \\ &\quad \times Y_{l'',m''}(\alpha, \beta) r^2 dr e^{-i\left(\frac{a}{2b}r^2 + \frac{d}{2b}\rho^2\right)} 4\pi \sum_{l''=0}^\infty \sum_{m''=-l''}^{l''} (i)^{l''} j_{l''}\left(\frac{r\rho_0}{b}\right) Y_{l'',m''}(\alpha_0, \beta_0) \\ &\quad \times \overline{Y_{l'',m''}(\varphi, \theta)} 4\pi \sum_{l=0}^\infty \sum_{m=-l}^l (i)^l j_l\left(\frac{\rho r}{b}\right) Y_{l,m}(\varphi, \theta) \overline{Y_{l,m}(\alpha, \beta)} \rho^2 \sin \alpha d\rho d\alpha d\beta. \end{aligned} \quad (34)$$

Using the orthonormality of the spherical harmonics and Eq. (30), we get

$$e^{\frac{1}{b}iu_0t}f(t) = \frac{8}{b^3} \sum_{l=0}^\infty \sum_{m=-l}^l (-i)^l Y_{l,m}(\varphi, \theta) \int_0^\infty f_{l,m}(r) S_{l,l}^{l,l}(\rho_0, \rho, \rho) \sum_{l'=0}^\infty \sum_{m'=-l'}^{l'} Y_{l',m'}(\varphi_0, \theta_0) \overline{Y_{l',m'}(\varphi, \theta)} \rho^2 d\rho, \quad (35)$$

which completes the proof. $\square$

4. Convolution theorems

Convolution theory plays a key role in the processing of signal, image, and optics, especially in the design and implementation of multiplicative filters. Various generalizations of classical convolution theory have been made to employ it in a wide range of fields [18,28–32]. In this section we discuss two types of convolution theorems of SLCT. The first type of convolution theorem is based on the special case of spherically symmetric functions. The second convolution theorem generalizes the results of the one-dimensional classical case.

4.1. Convolution theorem of spherically symmetric function

Definition 3. The convolution operation $\star_{SLCT}$ of the SLCT for two spherically symmetric functions $f(r)$ and $g(r)$ is defined as follows:

$$(f \star_{SLCT} g)(r) = \frac{2}{b^3 \sqrt{-2\pi ib}} e^{-i\frac{a}{2b}r^2} \int_0^\infty \int_0^\infty e^{i\frac{a}{2b}(r_0^2 + w^2)} g(r_0) f(w) S_0^{0,0}(w, r, r_0) w^2 r_0^2 dw dr_0, \quad (36)$$

where

$$S_0^{0,0}(w, r, r_0) = \int_0^\infty j_0\left(\frac{\rho w}{b}\right) j_0\left(\frac{\rho r_0}{b}\right) j_0\left(\frac{\rho r}{b}\right) \rho^2 d\rho. \quad (37)$$

Theorem 1. Let $h(r) = (f \star_{SLCT} g)(r)$. Then, there is a relation

$$\tilde{H}^A(\rho) = F^A(\rho) G^A(\rho), \quad (38)$$

where $\tilde{H}^A(\rho) = H^A(\rho) e^{-i\frac{d}{2b}\rho^2}$. $H^A(\rho)$, $F^A(\rho)$, and $G^A(\rho)$ denote the SLCT of the spherically symmetric functions $h(r)$, $f(r)$, and $g(r)$, respectively.

Proof. Only the $l = k = 0$ term is preserved in the series when the function is spherically symmetric, and according to Eq. (15), we obtain

$$Y_{0,0}(\varphi, \theta) = 1/\sqrt{4\pi}, \quad f_{0,0}(r) = f(r) \sqrt{4\pi}.$$

According to Eq. (30), we have

$$S_0^{0,0}(w, r, r_0) = \int_0^\infty j_0\left(\frac{\rho w}{b}\right) j_0\left(\frac{\rho r_0}{b}\right) j_0\left(\frac{\rho r}{b}\right) \rho^2 d\rho. \quad (39)$$

Hence

$$\begin{aligned}\tilde{h}(r) &= \frac{2}{\sqrt{-2\pi ib}} \frac{e^{-i\frac{d}{2b}\rho^2}}{b} \int_0^\infty e^{-i\frac{a}{2b}r^2} j_0\left(\frac{\rho r}{b}\right) \rho^2 d\rho \left\{ \frac{e^{i\frac{d}{2b}\rho^2}}{b} \int_0^\infty e^{i\frac{a}{2b}w^2} f(w) j_0\left(\frac{\rho w}{b}\right) w^2 dw \right\} \\ &\times \left\{ \frac{e^{i\frac{d}{2b}\rho^2}}{b} \int_0^\infty e^{i\frac{a}{2b}r_0^2} g(r_0) j_0\left(\frac{\rho r_0}{b}\right) r_0^2 dr_0 \right\} \\ &= \frac{2}{\sqrt{-2\pi ib}} \frac{e^{-i\frac{d}{2b}\rho^2}}{b} \int_0^\infty e^{-i\frac{a}{2b}r^2} \hat{F}_{0,0}^A(\rho) \hat{G}_{0,0}^A(\rho) j_0\left(\frac{\rho r}{b}\right) \rho^2 d\rho,\end{aligned}\quad (40)$$

where $\tilde{h}(r) = e^{i\frac{d}{2b}\rho^2} h(r)$, $\hat{F}_{0,0}^A(\rho)$ and $\hat{G}_{0,0}^A(\rho)$ is the SLCHT of $f(w)$ and $g(r_0)$, respectively.

Using the inverse transform of the SLCHT, and Eq. (19), we obtain

$$\tilde{H}^A(\rho) = F^A(\rho) G^A(\rho), \quad (41)$$

where $\tilde{H}^A(\rho) = e^{i\frac{d}{2b}\rho^2} H^A(\rho)$, which completes the proof. $\square$

Using the definition of the convolution operation $\star_{SLCT}$, the space of functions is shifted (breaking spherical symmetry) and then integrated over all the angular variables of the resulting shifted function. The unshifted function g is spherically symmetric and unaffected by integration over the angle variable [18].

Remark 3. When $A = (0, 1; -1, 0)$, the convolution operator in Definition 3 is equal to the convolution operator of the SFT in Ref. [18]. Theorem 1 degenerates into the classical convolution theory of an SFT with spherically symmetric functions.

4.2. Convolution theorem of non-spherically symmetric function

Definition 4. The convolution operation $\star_{SLCT}$ of the SLCT for two functions $f(r, \varphi, \theta)$ and $g(r, \varphi, \theta)$ is defined as follows:

$$\begin{aligned}(f \star_{SLCT} g)(r, \varphi, \theta) &= \frac{\sqrt{-2\pi ib}}{2(2l+1)} e^{-i\frac{a}{2b}(r^2-r_0^2)} \int_0^\infty \int_0^{2\pi} \int_0^\pi g(r_0, \varphi_0, \theta_0) \\ &\times F^{-1} [e^{-i\rho r_0 \cos \theta_0} F^A(b\rho, \alpha, \beta)](r, \varphi, \theta) r_0^2 \sin \varphi_0 dr_0 d\varphi_0 d\theta_0,\end{aligned}\quad (42)$$

where l is a real number, and $\cos \theta_0 = \sin \varphi_0 \sin \alpha \cos(\theta_0 - \beta) + \cos \varphi_0 \cos \alpha$.

Theorem 2. Let $h(r, \varphi, \theta) = (f \star_{SLCT} g)(r, \varphi, \theta)$, then

$$H^A(\rho, \alpha, \beta) = F^A(\rho, \alpha, \beta) G^A(\rho, \alpha, \beta), \quad (43)$$

where $H^A(\rho, \alpha, \beta)$, $F^A(\rho, \alpha, \beta)$, and $G^A(\rho, \alpha, \beta)$ denote the SLCT of $h(r, \varphi, \theta)$, $f(r, \varphi, \theta)$, and $g(r, \varphi, \theta)$, respectively.

Proof. From the spherical harmonic series expansion and inverse SFT, and by the popular formula [20], we have

$$\begin{aligned}&F^{-1} [e^{-i\rho r_0 \cos \theta_0} F^A(b\rho, \alpha, \beta)](r, \varphi, \theta) \\ &= \frac{1}{(2\pi)^3} \int_0^\infty \int_0^{2\pi} \int_0^\pi 4\pi \sum_{l''=0}^\infty \sum_{m''=-l''}^{l''} (-i)^{l''} j_{l''}(\rho r_0) \overline{Y_{l'',m''}(\varphi_0, \theta_0)} Y_{l'',m''}(\alpha, \beta) \frac{2}{\sqrt{-2\pi ib}} \sum_{l'=0}^\infty \sum_{m'=-l'}^{l'} (-i)^{l'} \\ &\times \hat{F}_{l',m'}^A(b\rho) Y_{l',m'}(\alpha, \beta) 4\pi \sum_{l=0}^\infty \sum_{m=-l}^l (i)^l j_l(\rho r) Y_{l,m}(\varphi, \theta) \overline{Y_{l,m}(\alpha, \beta)} \rho^2 \sin \alpha d\alpha d\beta.\end{aligned}\quad (44)$$

According to the Slater coefficients [18,24,25], Eq. (44) becomes

$$\begin{aligned}&\frac{4}{\pi \sqrt{-2\pi ib}} \int_0^\infty \sum_{l''=|l-l'|}^{l+l'} c^{l''}(l, m, l', m') j_{l''}(\rho r_0) \overline{Y_{l'',m-m'}(\varphi_0, \theta_0)} \sum_{l'=0}^\infty \sum_{m'=-l'}^{l'} (-i)^{l'+l''} \hat{F}_{l',m'}^A(b\rho) \\ &\times \sum_{l=0}^\infty \sum_{m=-l}^l (i)^l j_l(\rho r) Y_{l,m}(\varphi, \theta) \rho^2 d\rho.\end{aligned}\quad (45)$$

Using the fact that

$$\sum_{m=-l}^l c^{l''}(l, m, l', m') = (2l+1) \delta_{l'',0} = \begin{cases} 2l+1, & l''=0 \\ 0, & \text{otherwise} \end{cases}, \quad (46)$$

where $\delta_{i,j}$ denotes the kronecker delta. For $l = l'$, $\tilde{g}(r_0, \varphi_0, \theta_0)$ can be expanded in a spherical Fourier series, Eq. (45) becomes

$$\tilde{h}(r, \varphi, \theta) = \frac{2}{\pi} \sum_{l=0}^\infty \sum_{m=-l}^l \underbrace{\int_0^\infty \tilde{g}_{l,m}(r_0) j_l(\rho r_0) r_0^2 dr_0}_{\hat{G}_{l,m}^A(\rho)} \int_0^\infty \hat{F}_{l,m}^A(b\rho) j_l(\rho r) Y_{l,m}(\varphi, \theta) \rho^2 d\rho, \quad (47)$$

where $\tilde{g}_{l,m}(r_0) = e^{i\frac{a}{2b}r_0^2} g_{l,m}(r_0)$, and $\hat{G}_{l,m}(\rho)$ is the l th order SHT of $\tilde{g}_{l,m}(r)$. Therefore

$$\tilde{h}_{l,m}(r) = \frac{2}{\pi} \int_0^\infty \hat{G}_{l,m}(\rho) \hat{F}_{l,m}^A(b\rho) j_l(\rho r) \rho^2 d\rho, \quad (48)$$

where $\tilde{h}_{l,m}(r) = e^{i\frac{a}{2b}r^2} h_{l,m}(r)$. Using of the SHT's inversion formula, aconvolution relation is obtained

$$H^A(\rho, \alpha, \beta) = F^A(\rho, \alpha, \beta) G^A(\rho, \alpha, \beta), \quad (49)$$

which completes the proof. $\square$

Remark 4. When $A = (0, 1; -1, 0)$, [Theorem 2](#) degenerates into the classical convolution theory of the SFT with non-spherical symmetric functions.

5. Potential applications

The first potential application is in the field of acoustics. In three-dimensional audio and virtual auditory space, sound source positioning technology obtains the spatial position of the target sound source through a microphone array. Recently, locating three-dimensional sound sources based on SFT has received considerable attention [\[12\]](#). This method reduces the complexity of the calculation without increasing the number and volume of the device. Because of the limitations of the traditional SFT in processing non-stationary signals in complex scenarios, the SLCT has four additional parameters, which can obtain detailed information about the three-dimensional azimuth of the sound source. The following is a sound-field analysis based on acoustic principles. When a plane wave is incident at an angle of (φ_0, θ_0) , it is decomposed in the spherical coordinate system, and the sound pressure formed by the incident wave is given by [\[12\]](#):

$$S_{in}(\varphi, \theta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l 4\pi(i)^l j_l(k\lambda) D(\varphi_0, \theta_0) Y_{l,m}(\varphi, \theta), \quad (50)$$

where $D(\varphi_0, \theta_0) = \overline{Y_{l,m}(\varphi_0, \theta_0)}$ is the orientation factor, which is a function of the incidence angle. λ is the radius of the rigid sphere, (φ_0, θ_0) is the spherical coordinate form of the plane wave incidence angle, and $j_l(k\lambda)$ is a spherical Bessel function of order l .

Because the surface of the ball is made of a hard shell, sound waves are reflected. The sound field formed by reflection is:

$$S_{re}(\varphi, \theta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l -4\pi(i)^l \frac{j'_l(k\lambda)}{h'_l(2)(k\lambda)} h_l^{(2)}(k\lambda) D(\varphi_0, \theta_0) Y_{l,m}(\varphi, \theta), \quad (51)$$

where $h_l^{(2)}(k\lambda)$ is the ball-type Hankel function; $j'_l(k\lambda)$, and $h_l'^{(2)}(k\lambda)$ are the first derivatives of the ball-type Bessel function and ball-type Hankel function, respectively.

Hence, the total sound pressure is:

$$S(\varphi, \theta) = S_{in}(\varphi, \theta) + S_{re}(\varphi, \theta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l b_l(kr, k\lambda) D(\varphi_0, \theta_0) Y_{l,m}(\varphi, \theta), \quad (52)$$

where $b_l(kr, k\lambda) = 4\pi i^l \left(j_l(kr) - \frac{j'_l(k\lambda)}{h_l'^{(2)}(k\lambda)} h_l^{(2)}(k\lambda) \right)$ represents the characteristic amplitude. It is noteworthy that the first sum of Eq. (52) indicates that l tends towards infinity, which is impossible to achieve in practice. Eq. (52) should be truncated. When N order is taken, the sound pressure of the spherical surface $S_N(\varphi_j, \theta_j)$ can be expressed as:

$$S_N(\varphi_j, \theta_j) = \sum_{l=0}^N \sum_{m=-l}^l b_l(kr) D(\varphi_0, \theta_0) Y_{l,m}(\varphi_j, \theta_j). \quad (53)$$

The transform pair can be written as:

$$S_{lm}(\varphi_j, \theta_j) = \sum_{j=0}^L v_j S_N(\varphi_j, \theta_j) \overline{Y_{l,m}(\varphi_j, \theta_j)}, \quad (54)$$

where N is the order of sampling, L is the number of sampling microphones on the sphere, and v_j is the compensation-weighted value of the j th microphone.

Based on the above analysis, the sound field is described as a linear combination of various spherical harmonics, and the sound field is transformed by SLCT. The SLCT of source amplitude density can be obtained owing to the orthogonal characteristics of the spherical harmonics. Through inverse SLCT, the amplitude density of each sound source can be obtained, and the spatial position of the sound source can be determined. These theoretical results suggest applications of SLCT in locating a three-dimensional sound source. In the future, our intend to experimentally verify these theories.

Potential applications include optical measurements. SFT-based point-based diffraction interference involves time-frequency domain conversion and filtering, and the phase information of the light source can be obtained. The frequency domain in SFT optics may be a special location in the transformation domain in SLCT optics; that is, each characteristic and every application in the SFT is a special case of the SLCT. Therefore, it is possible to promote and improve the use of SLCT in every field that employs SFT, providing a wide scope for the development of SLCT optics.

6. Conclusions

Compared with SFT, the proposed SLCT has greater flexibility in processing non-stationary signals. First, the definitions of SLCT and SLCHT in spherical polar coordinates are provided. As a preliminary result, the relationship between SLCT and SLCHT is obtained based on the orthogonality of the spherical harmonics. Some basic properties and two convolution theorems of the SLCT are derived using this relation. The first convolution theorem is based on the special case of spherically symmetric functions. The second convolution theorem generalizes the corresponding one-dimensional case results. Finally, some potential applications of SLCT are proposed. In the future, our research efforts will focus on the practical applications of SLCT.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article

Acknowledgments

This work was supported by the National Natural Science Foundation of China [No. 62171041].

References

- [1] X.G. Xia, On bandlimited signal with fractional Fourier transform, *IEEE Signal Process. Lett.* 3 (3) (1996) 72–74.
- [2] D.Y. Wei, Y.M. Li, Generalized sampling expansions with multiple sampling rates for lowpass and bandpass signals in the fractional Fourier transform domain, *IEEE Trans. Signal Process.* 64 (18) (2016) 4861–4874.
- [3] J. Shi, J. Zheng, X. Liu, W. Xiang, Q. Zhang, Novel short-time fractional Fourier transform: theory, implementation, and applications, *IEEE Trans. Signal Process.* 68 (2020) 3280–3295.
- [4] D.Y. Wei, Y.J. Zhang, Fractional stockwell transform: Theory and applications, *Digit. Signal Process.* 115 (2021) 103090, 1–20.
- [5] D.Y. Wei, Y.J. Zhang, Y.M. Li, Linear Canonical stockwell transform: Theory and applications, *IEEE Trans. Signal Process.* 70 (2022) 1333–1347.
- [6] J. Meyer, G. Elko, Theory and design of high order sound field microphones using spherical microphone array, in: *IEEE International Conference on Audio Speech and Signal Processing, ICASSP, Orlando, Florida, USA, 2002*, pp. 1781–1784.
- [7] T.D. Abhayapala, D.B. Ward, Unified Wigner-Ville distribution and ambiguity function in the linear Canonical transform domain, in: *IEEE International Conference on Audio Speech and Signal Processing, ICASSP, Orlando, Florida, USA, 2002*, pp. 1949–1952.
- [8] S.F. Yan, et al., Optimal modal beamforming for spherical microphone arrays, *IEEE Trans. Audio Speech Lang. Process.* 19 (2) (2011) 361–371.
- [9] M. Xu, L.V. Wang, Time-domain reconstruction for thermoacoustic tomography in a spherical geometry, *IEEE Trans. Med. Imaging.* 21 (7) (2002) 814–822.
- [10] J.R. Driscoll, D.M. Healy, Computing Fourier transforms and convolutions on the 2-sphere, *Adv. Appl. Math.* 15 (1994) 202–250.
- [11] R. Basri, D. Jacobs, Lambertian reflectance and linear subspaces, in: *Proceedings of the Eighth IEEE International Conference on Computer Vision, ICCV, Vancouver, BC, Canada, 2001*, pp. 383–390.
- [12] Y.Q. Tang, Q.H. Huang, Y. Fang, Localization of sound sources based on spherical Fourier transformation in three-dimension, *Signal Process. (in Chinese)* 26 (2010) 654–658.
- [13] L.Y. Zhang, et al., Sound source localization using uon-conformal surface sound field transformation based on spherical harmonic wave decomposition, *Sensors* 17 (5) (2017) 1087, 1–12.
- [14] N. Baddour, Fourier diffraction theorem for diffusion-based thermal tomography, *J. Phys. A* 39 (2006) 14379–14395.
- [15] Y.N. Sun, B.Z. Li, Sliding discrete linear Canonical transform, *IEEE Trans. Signal Process.* 66 (17) (2018) 4553–4563.
- [16] S.Q. Xu, L. Feng, Y. Chai, Y.G. He, Analysis of A-stationary random signals in the linear Canonical transform domain, *Signal Process.* 146 (2018) 126–132.
- [17] G.S. Chirikjian, A.B. Kyatkin, *Orthogonal expansions in curvilinear coordinates*, in: *Engineering Applications of Noncommutative Harmonic Analysis*, CRC, Boca Raton, Florida, USA, 2000, pp. 99–102.
- [18] N. Baddour, Operational and convolution properties of three-dimensional Fourier transforms in polar coordinates, *J. Opt. Soc. Am. A* 27 (10) (2010) 2144–2155.
- [19] B. kursunoglu, Angular momentum, in: *Modern Quantum Theory*, Freeman, San Francisco, London, UK, 1962, pp. 114–115.
- [20] N. Hahn, F. Schultz, S. Spors, Time domain sampling of the radial functions in spherical harmonics expansions, *IEEE Trans. Signal Process.* 69 (2021) 4502–4512.
- [21] H. Ikeno, Spherical Bessel transform via exponential sum approximation of spherical Bessel function, *J. Comput. Phys.* 355 (2018) 426–435.
- [22] R. Piessens, Hankel transform, in: *Transform and Applications Handbook*, CRC, Boca Raton, Florida, USA, 2000, pp. 9.1–9.3.
- [23] A.I. Zayed, Sampling of signals bandlimited to a disc in the linear Canonical transform, *IEEE Signal Process. Lett.* 25 (2018) 1765–1769.
- [24] J.C. Slater, The elementary of multiplets, in: *Quantum Theory of Atomic Structure*, McGraw-Hill, New York, USA, 1960, pp. 309–311.
- [25] S. Khenchoul, et al., Analytical evaluation of two-center integrals for Slater-orbitals type in hartree-fock formalism on noncommutative space, *Int. J. Theor. Phys.* 58 (2019) 1071–1078.
- [26] R. Mehrem, J.T. Londergan, M.H. Macfarlane, Analytic expressions for integrals of products of spherical Bessel functions, *J. Phys. A Math. Gen.* 24 (1991) 1435–1453.
- [27] V. Fabrikant, Computation of infinite integrals involving three Bessel functions by introduction of new formalism, *Z. Angew. Math. Mech.* 83 (6) (2003) 363–374.
- [28] W.B. Gao, B.Z. Li, Convolution theorem involving n-dimensional windowed fractional Fourier transform, *Sci. China Inf. Sci.* 64 (2021) 169302, 1–3.
- [29] Z.C. Zhang, Convolution theorem for two-dimensional LCT of angularly periodic functions in polar coordinates, *IEEE Signal Process. Lett.* 26 (8) (2019) 1142–1146.
- [30] B. Deng, R. Tao, Y. Wang, Convolution theorems for the linear Canonical transform and their applications, *Sci. China Inf. Sci.* 49 (2006) 592–603.
- [31] D.Y. Wei, Convolution and multichannel sampling for the offset linear Canonical transform and their applications, *IEEE Trans. Signal Process.* 67 (23) (2019) 6009–6024.
- [32] J. Shi, X.P. Liu, N.T. Zhang, Generalized convolution and product theorems associated with linear Canonical transform, *Signal Image Video Process.* 8 (2014) 967–997.